4. Data Generation and Preparation

Chapter 3 fixed the plant. It did not fix how its two simulated representations meet. The Simscape model knows the hydraulics but is far too slow to train a policy against, and the IsaacLab articulation is fast but has no hydraulics at all. Everything the learned models of Chapter 5 can say about the actuator has to travel from the first world to the second inside a dataset. This chapter describes how those datasets were designed, collected, audited and prepared.

The datasets come from two different sources, and keeping their provenance straight is a precondition for reading the results honestly (Table 3.3).

Historical Simscape provenance. Two Simscape campaigns identified the frozen actuator models that this thesis carries forward as comparison components: the command-conditioned reduced-order model and the inverse/strict-forward pair. The first campaign drives the arm with a closed-loop controller and produces the realistic operating distribution the inverse and reduced-order models learn from. The second removes the controller entirely and applies deliberate excitation to the valves, because the closed-loop campaign turned out to be unable, in principle, to identify what the valve does. Section 4.4 sets out that diagnosis and its repair in full, because the failure is easy to repeat and the repair is what makes the forward model usable.

Closed-loop calibration of the ROM in IsaacLab. The command ROM is not a pure Simscape artifact, whereas its partner in the comparison, the ID–FD pair, *is*. After Simscape training the ROM was fine-tuned on closed-loop rollouts collected inside IsaacLab itself, on the five plant-reachable reference shapes, with only a small bounded correction head left trainable and the Simscape material retained as a replay term (Section 4.6). It is therefore the one actuator model in this thesis that has seen the plant it is deployed in.

The IsaacLab adaptation collection. The system-dynamics surrogate, the privileged factor encoder, and the Arm Adaptation Module are not trained on Simscape data at all. They are trained on a 118-run collection made in IsaacLab through the deployment plant itself (the command ROM plus PhysX), driven by the DLS controller with dedicated PRBS excitation windows (Section 4.7). The reason is simple: those models exist to describe the plant the adaptive controller faces, and that plant is the IsaacLab one. No new Simscape data was collected for this thesis, and no earlier adaptation or system-dynamics model